

RUSHI ASHOK SAVANI

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WORK EXPERIENCE

AI Engineer / Data Science Intern

Jun 2025 – May 2026

ProHealth Home Health and Hospice

Atlanta GA

- Built an agentic text-to-SQL pipeline on AWS using LangGraph, MCP tooling, DynamoDB for session memory and self-correcting loops, cutting administrator lookup time by 95%.
- Evaluate the agentic pipeline on a 100+ question set through Postman runs with RAGAS and human review, fixing retrieval inconsistency with AWS Opensearch hybrid retrieval and reranking to reach 92.6% context precision and 98% faithfulness.
- Retrained a readmission ML model on AWS Sagemaker with severity ranked ordinal over one-hot encoding and an F2 optimized threshold, lifting recall by 30% and cutting reamission cases by 10% (~\$15K savings).
- Architected three LangChain pipelines with parameterized SQL and deterministic chunking across predefined data points, giving finance and admin teams reproducible, auditable root-cause reports using LLMs, surfacing about \$80K annually.
- Validated compliance, hallucination and prompt versions in LangSmith against a de-identified 150-patient dataset using LLM-as-the-judge and clinician review, making model and prompt testing faster and traceable.
- Shipped the LLM and ML pipeline as a FastAPI feature inside the SaaS product, deploying SageMaker on ECS via peer-reviewed GitHub Actions CI/CD, fixing a 10% recall drift on CloudWatch.

Data Science Research Assistant

Jan 2025 – May 2025

Syracuse University

Syracuse, NY

- Fine-tuned a RoBERTa sentiment model on 7,000+ survey responses using PyTorch and Hugging Face, adjusting class weights with stratified k-fold validation to improve labeling accuracy by 30%.
- Automated a data pipeline for survey-response cleaning across missing values, mismatched types, and multilingual text, saving 15+ hours of manual work.
- Clustered survey responses with Sentence-BERT embeddings and HDBSCAN to surface AI-adoption patterns, showing insights through Power BI dashboards with daily auto-refreshes.

Data Analyst

Jan 2023 - Jun 2024

Krishna Jewels

Mumbai, IN

- Forecasted seasonal demand and pricing trends with time-series analysis in Excel (Solver, Power Query, Power Pivot), informing pricing decisions that grew revenue 8-10% year-over-year.
- Designed and indexed a SQL Server database schema for fast multi-table KPI joins, cutting manager lookup time by 95%.
- Led a 4-member team to complete migrating 40K+ records from paper and Excel into SQL database with 98% data accuracy.
- Communicated monthly on branch-level insights through interactive Power BI dashboards to executives, building 10+ KPIs and driving 40% improvement in stock allocation across 4 branches.

TECHNICAL SKILLS

Languages & Dev Tools: Python, SQL, TypeScript, Node.js, Claude Code, Cursor, Codex

Frameworks: PyTorch, TensorFlow, Hugging Face Transformer, FastAPI, LangChain/LangGraph, Google ADK, LlamaIndex

ML & Deep Learning: Scikit-learn, Pandas, NumPy, XGBoost, Gradient Boosting, Logistic Regression, Decision Tree, SVM, Bayesian Hyperparameter Tuning, Classification Metrics, A/B Testing, Natural Language Processing (NLP), Sentiment Analysis, BERT modeling

Generative AI: RAG, MCP, LLMs, Prompt Engineering, LiteLLM Inference, Evaluation Frameworks (RAGAS, DeepEval, LangSmith)

MLOps & Cloud: AWS SageMaker, Docker, GitHub Actions CI/CD, AWS (EC2, ECS, S3), CloudWatch

Databases: SQL Server, Snowflake, Vector Store (pgvector), AWS Redshift

BI & Analytics: Power BI, Tableau, Streamlit, ETL/ELT, Data Modeling, Statistical Analysis, Hypothesis Testing

EDUCATION

Syracuse University, New York, USA

Aug 2024 - May 2026

Masters | Information Systems, Data Science

(GPA: 3.86/4)

- Relevant Coursework:* Applied Machine Learning, Database Management, Data Science, Cloud Management

Thadomal Shahani Engineering College, Mumbai, India

Aug 2020 - Jun 2024

Bachelors | Information Technology

(GPA: 8.52/10)

- Relevant Coursework:* Artificial Intelligence and Data Science, Software Engineering, DevOps, Cloud Computing

PROJECTS

PrivateRAG | *Fully Local Private-Document Q&A System*, [GitHub](#)

- Designed a fully local **Retrieval-Augmented Generation (RAG)** stack with document ingestion, Qdrant vector indexing, metadata filtering, and **Llama 3.1** inference with zero external API calls.
- Wrote a **TypeScript** chat interface over a local OpenAI-compatible backend with streaming responses, source attribution, and chunk-level traceability.

AI Email Response System | *Autonomous Emailing Pipeline*, [GitHub](#)

- Cut response wait time 90% by engineering a RAG email-reply generator that checks escalation rules first, drafting a hand-off instead of an autonomous resolution for high-risk cases.
- Reduced human touch on inbound support email by auto-routing policy-clean, high-confidence replies through a provider-agnostic MCP client, escalating only cases a ground-truth-validated judge flags as policy-mandated.

COVID-19 India Analytics | *Interactive Tableau Dashboard*, [Dashboard](#)

- Implemented a **Python** pipeline to reconcile inconsistent state-name spellings across public datasets, powering an interactive **Tableau** dashboard that flagged states with high testing but low vaccination coverage.